

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech (EE) Examination

March-April 2018

EE404 Energy Management and Conservation

Date: 02.04.2018, Monday

Time: 10.00 a.m. To 01.00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the question below. [07]

- a. Define various resources of energy with examples.

OR

- a. Define laws of conservation of energy and explain with example of energy conversion process.

Q – 2.a How to Improve power factor and what are the impacts of improved power factor in electrical systems. [04]

Q – 2.b Answer any two. [10]

- (i) Explain Energy Conservation Act-2001 and discuss the role plays of BEE, State Government and central government. [05]
- (ii) How we can use Electric motors in efficient way? Or Explain Energy savings in Electric Motors. [05]
- (iii) Explain the Latent Heat or Phase Change Energy Storage System in detail. [05]

Q - 3 Answer any two. [14]

- a. Explain Seven Principles of Energy Management. [07]
- b. Explain Transformer Load Management with example. [07]
- c. Draw Classification chart of furnaces and write characteristic of an efficient Furnace. [07]

SECTION – II

Q - 4 Answer the question below. [07]

- a. What is Energy Management? Explains steps involved in EM.

OR

- a. Explain Indoor and outdoor Lighting Configuration.

Q – 5.a Explain need of cogeneration and principle of cogeneration. **[05]**

OR

Q – 5.a Write down the equipments and instruments with their applications used for energy audit process. **[05]**

Q – 5.b Write the Potential Energy Conservation Opportunities in Motor and Transformer. **[04]**

Q – 5.c Explain the Pumped Hydroelectric (PHS) and Compressed Air Energy Storage (CAES) System in detail. **[05]**

OR

Q – 5.c Explain the Super Conducting Magnetic Energy Storage (SMES) System in detail. **[05]**

Q – 6.a Explain the Double Layer Capacitor or Super Capacitor Energy Storage Systems in detail. **[07]**

Q – 6.b Explain the Detail Energy Audit with ten steps methodology. **[07]**

OR

Q – 6.b During April-2017, the plant has recorded maximum demand 600 kVA, has average power factor is 0.82 lag, the minimum average power factor to be maintained is 0.92 lag and every 1% difference in power factor attracts penalty of Rs 10,000 in each month. **[07]**

- a) Calculate the improvement in power factor for May 2017 by installing 100 kVAR capacitors.
- b) Calculate the penalty to be paid if any during May 2017.
